

Cosmic-muon trigger and coincidence rates in the aligned ESA test stand

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Abstract

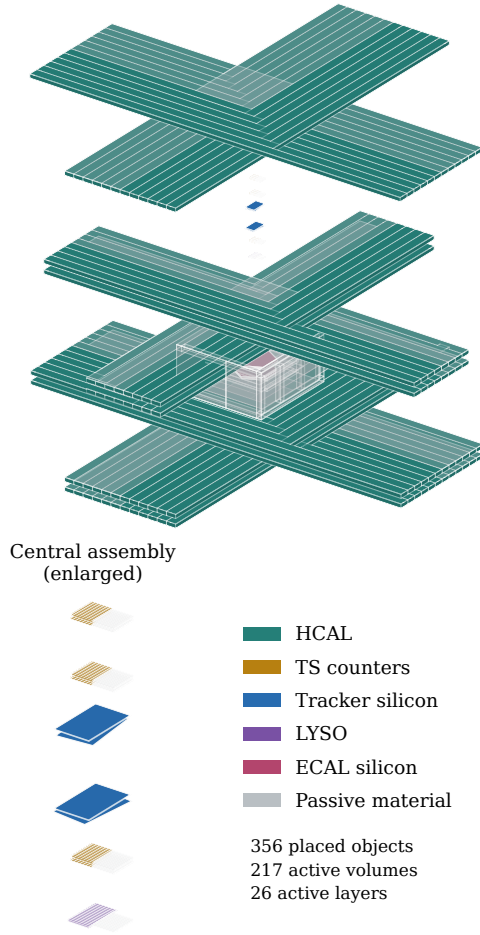
We estimate cosmic-muon coincidence rates in the ESA test stand with the ECAL centred on the tracker axis and the remaining detectors at their original heights. A golden cosmic event crosses all 26 active layers. Its estimated rate is 2.163 per hour without an overhead slab and 1.842 per hour beneath 3 ft of concrete. Requiring only the ten HCAL layers gives 13,260 and 11,290 per hour, respectively. We compare HCAL trigger definitions, relax the tracker and other detector requirements, and tabulate every combination of the three HCAL stations, the TS, LYSO and ECAL requirements, and tracker multiplicity. The calculation uses the registered detector geometry, a sea-level muon spectrum and continuous-range estimates along each trajectory. These are muon-crossing rates before detector response; measured trigger rates will also depend on thresholds, efficiency, noise and live time.

Overview

The stand combines a compact trigger-scintillator (TS), silicon-tracker and LYSO assembly with an ECAL module and three much larger HCAL stations. The HCAL spans metres, while the TS bars are only 30 mm wide in X . A muon crossing all the central devices must therefore pass through a narrow common aperture. Figures 1 and 2 show how this aperture controls the rate. This difference sets the scale of the commissioning samples: an HCAL coincidence collects many muons, while a golden event connects every detector through a single track.

For choosing a trigger, Table 2 gives the rates for different HCAL coincidences. For planning measurements with the central devices, Table 3 gives the rate retained by each added requirement. Both tables compare no overhead slab with the assumed 3 ft concrete slab and quote rates before detection losses. The full set of combinations is collected in Appendix A; the geometry, flux and stopping assumptions are described in Sections 1 and 4.

Aligned geometry used for the rates



Earlier offset layout: accepted rays

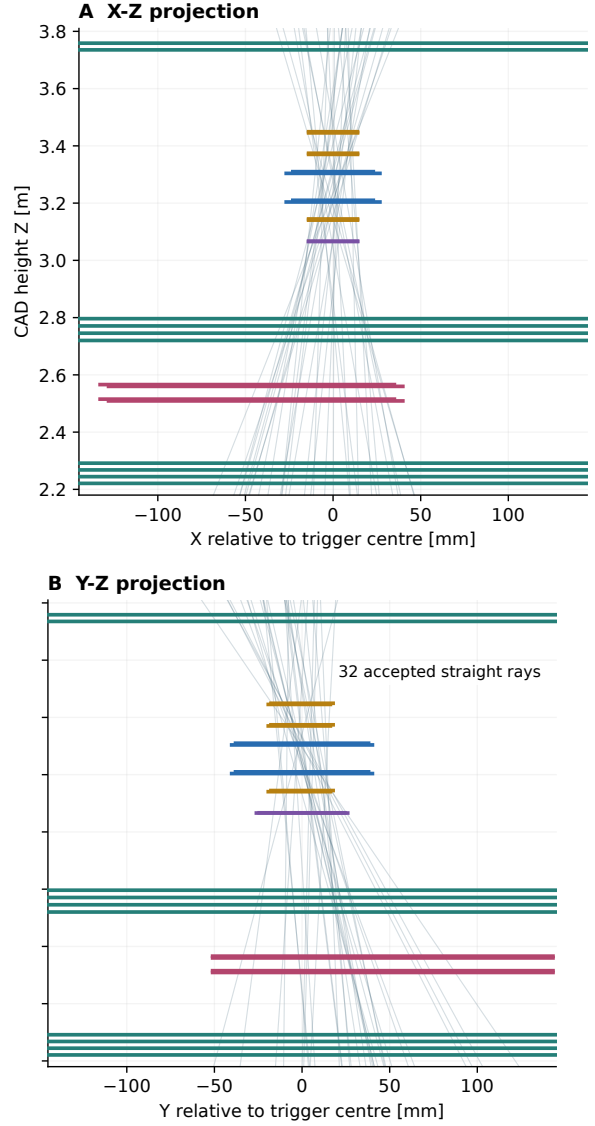
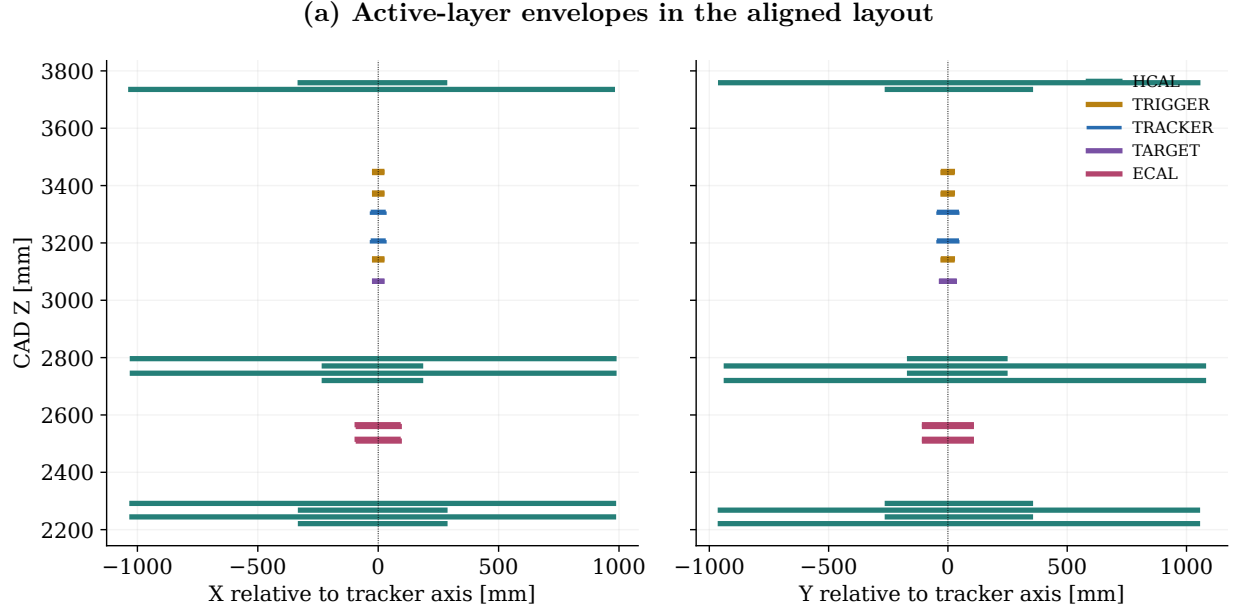


Figure 1: **Installed geometry and the accepted-ray aperture.** Left: the complete aligned detector model, with passive material shown translucently and a separate enlargement of the central assembly below. CAD stand supports are outside the material model. Right: the previous rate report's 32 accepted straight rays, with its original vector projections stacked for readability. These rays use the earlier *offset* ECAL; all main-table rates use the aligned layout. Colours identify the same subsystems in both panels. The rays include no scattering or detector response.



(b) Rates retained by representative detector requirements

Parentheses: no roof / 3 ft concrete, in muons per hour

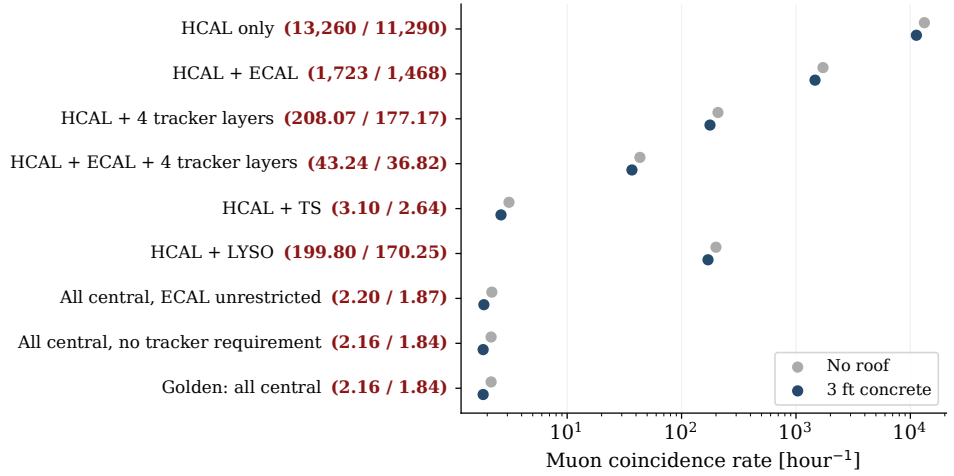


Figure 2: **From detector aperture to coincidence rate.** (a) The active-layer envelopes, with transverse coordinates measured from the tracker axis and heights in CAD Z . Line thicknesses are enlarged for visibility; the calculation retains the segmented solids. (b) Inclusive selections requiring all ten HCAL layers, on a logarithmic rate axis. “All central” requires all TS, LYSO, ECAL and tracker layers. “ECAL unrestricted” keeps ECAL installed but imposes no ECAL hit requirement: events may hit or miss it, while all other listed hit requirements still apply. “No tracker requirement” similarly leaves tracker hits unrestricted. The same aligned geometry and sea-level spectrum are used for both roof assumptions. Bold red values beside each selection give the rate per hour (no roof / 3 ft concrete).

1 Reference layout and event definitions

The reference configuration moves the entire ECAL by $\Delta X = +46.5596$ mm and $\Delta Y = -46.0763$ mm in the CAD coordinate system. Its combined active-silicon envelope is then centred on the tracker/TS/LYSO axis at $(X, Y) = (5027.51999, -155.75023)$ mm. The four ECAL silicon layers retain their internal offsets. No detector moves vertically, and all three HCAL stations remain in place. This is the previously studied “centred only” layout; the raised assembly is retained as a historical comparison in Appendix C.

Table 1: Active detector inventory. Heights are CAD Z ; within each range, layers are ordered from top to bottom.

Requirement	Layers	Active volumes	Height range (mm)
H_T : top HCAL	2	24	3758.711–3735.339
S : TS counters	6	72	3449.388–3140.588
T_1, T_2, T_3, T_4 : tracker	4	4	3309.888–3203.888
Y : LYSO	2	33	3066.788–3065.988
H_M : HCAL above ECAL	4	32	2796.346–2720.146
E : ECAL silicon	4	4	2566.032–2508.962
H_B : HCAL below ECAL	4	48	2291.504–2220.931
Golden event	26	217	All requirements

We call an event *golden* when one downward-going muon crosses an active volume in every layer in Table 1. In a segmented layer, crossing any one bar is sufficient. A hit requires a positive path length in the active volume; it does not require passage through the layer midplane. The inactive silicon borders do not count as tracker hits.

Throughout this note, “everything minus ECAL” means that ECAL hits are unrestricted. The ECAL and its passive material remain installed. Likewise, “minus one tracking layer” means at least three of the four tracker layers, with any one layer allowed to lack a hit. It includes four-layer events. These inclusive samples overlap and must not be added. The separate `exclusive_tracker_counts.csv` table gives exact tracker multiplicities; physically removing equipment is a different geometry study.

The three HCAL stations are denoted H_T (top, two layers), H_M (above ECAL, four layers) and H_B (below ECAL, four layers). Unless explicitly stated otherwise, requiring a station means requiring *all* its active layers. The TS requirement S means all six trigger-scintillator layers; Y means both LYSO layers; E means all four ECAL silicon layers. A tracker multiplicity $T_{\geq k}$ accepts any set of at least k of the four active tracker layers.

2 HCAL trigger rates

Table 2 compares strict layer coincidences and a looser alternative that accepts at least one layer in each selected HCAL station. An entry such as $H_T H_B$ is an AND between the two stations, with the middle station unrestricted. The “any station” and “at least two stations” rows are unions: each muon is counted once even if it satisfies several station choices. No TS, tracker, LYSO or ECAL hit is required in this table.

We use the strict ten-layer coincidence as the reference sample for the equipment comparisons below. Beneath the modelled concrete slab, the golden sample is 0.0163% of this coincidence sample,

Table 2: HCAL muon-trigger estimates per hour. Strict means every layer in each selected station; loose means any layer in each selected station. A suffix k means 10^3 events/hour. These rows impose no central-detector requirements; the full numerical errors are in the CSV tables.

Stations	Strict: all station layers		Loose: any station layer	
	No roof	Concrete	No roof	Concrete
H_T	205.8k	175.4k	1154.6k	982.3k
H_M	81.3k	69.5k	874.1k	744.7k
H_B	187.8k	160.7k	1198.7k	1022.5k
H_TH_M	16.5k	14.0k	374.3k	317.7k
H_TH_B	17.7k	15.1k	345.7k	293.6k
H_MH_B	38.1k	32.5k	582.4k	496.4k
$H_TH_MH_B$	13.3k	11.3k	274.1k	233.0k
Any station	415.9k	355.4k	2199.2k	1874.9k
At least two stations	45.8k	39.0k	754.1k	641.7k

or about one golden event per 6,128 strict HCAL coincidences. The corresponding mean golden-event interval is 32.6 minutes, before detection losses. A broader HCAL trigger therefore provides a much larger sample for detector commissioning, although most of those muons miss the small central devices.

These calculations describe muons passing the stated logical conditions. They are not a prediction of discriminator singles or accidental coincidences. The coincidence time window and measured channel rates are needed to estimate accidentals. The event logic also needs to be implemented consistently in hardware or offline selection: an OR over bars within a layer is different from an OR over layers within a station.

3 Rates for detector combinations

Table 3 collects the requested golden-event relaxations and several useful commissioning selections. Every row requires the strict $H_TH_MH_B$ coincidence. Rates are given per hour; the quoted errors are numerical integration errors only.

Within the numerical precision of this study, allowing a tracker layer to be missed does not increase the rate once both the TS and LYSO coincidences are required. Those small detectors already confine accepted trajectories to the tracker footprint. This does not imply perfect tracker efficiency. A relaxed layer requirement can still recover events lost to inactive channels or detection inefficiency, neither of which is included in this calculation. Ignoring ECAL hits gives a small additional acceptance because the aligned ECAL already covers nearly all of this narrow trajectory sample.

Tracker requirements matter more when the TS or LYSO requirements are removed. Tracker and ECAL acceptance then select a much smaller subset of the HCAL trigger sample, and the individual tracker choices can have different rates. Table 4 makes the single-layer omissions explicit.

Appendix A contains the complete 280-row comparison, displayed as two matrices: seven nonempty HCAL station choices, eight choices of S, Y, E , and five tracker thresholds. The accompanying machine-readable tables contain all 896 specified tracker-subset selections as well as the inclusive and exclusive multiplicities. This enumerates the stated subsystem and tracker choices; arbitrary masks of the 26 individual layers are outside its scope. Individual HCAL-layer singles are tabulated separately in Appendix B.

Table 3: Requested selections, in muons/hour, with $H = H_T H_M H_B$. An omitted hit requirement removes no material. Errors are numerical standard errors only. Each tracker omission is inclusive.

Selection	No roof	3 ft concrete
Golden: everything	2.16319 ± 0.00063	1.84240 ± 0.00054
Everything minus ECAL	2.19533 ± 0.00066	1.86976 ± 0.00056
Everything minus one tracker layer	2.16319 ± 0.00063	1.84240 ± 0.00054
Minus ECAL and one tracker layer	2.19533 ± 0.00066	1.86976 ± 0.00056
Everything minus two tracker layers	2.16319 ± 0.00063	1.84240 ± 0.00054
Minus ECAL and two tracker layers	2.19533 ± 0.00066	1.86976 ± 0.00056
Everything minus three tracker layers	2.16319 ± 0.00063	1.84240 ± 0.00054
Minus ECAL and three tracker layers	2.19533 ± 0.00066	1.86976 ± 0.00056
Everything minus all tracker layers	2.16319 ± 0.00063	1.84240 ± 0.00054
$H + S + Y$; no ECAL or tracker cut	2.19533 ± 0.00066	1.86976 ± 0.00056
$H + Y + E + T_{\geq 4}$; no TS cut	19.4408 ± 0.0047	16.5575 ± 0.0040
$H + E + T_{\geq 4}$	43.2409 ± 0.0036	36.8237 ± 0.0030
$H + Y + T_{\geq 4}$	28.7179 ± 0.0030	24.4563 ± 0.0026
$H + E$	$1,722.57 \pm 0.16$	$1,467.58 \pm 0.14$
$H + T_{\geq 4}$	208.068 ± 0.020	177.174 ± 0.017
$H + S$	3.09906 ± 0.00048	2.63939 ± 0.00041
$H + Y$	199.805 ± 0.036	170.248 ± 0.030
H only: all ten HCAL layers	$13,259.6 \pm 2.8$	$11,290.2 \pm 2.4$

Table 4: Specified single-tracker omissions with H, S, Y and the other three tracker layers required. The omitted layer may still be hit. Tracker heights are $Z = (3309.888, 3303.888, 3209.888, 3203.888)$ mm for T_1 through T_4 .

Unrestricted tracker	ECAL	No roof/hour	Concrete/hour
T_1	Required	2.16319 ± 0.00063	1.84240 ± 0.00054
T_1	Unrestricted	2.19533 ± 0.00066	1.86976 ± 0.00056
T_2	Required	2.16319 ± 0.00063	1.84240 ± 0.00054
T_2	Unrestricted	2.19533 ± 0.00066	1.86976 ± 0.00056
T_3	Required	2.16319 ± 0.00063	1.84240 ± 0.00054
T_3	Unrestricted	2.19533 ± 0.00066	1.86976 ± 0.00056
T_4	Required	2.16319 ± 0.00063	1.84240 ± 0.00054
T_4	Unrestricted	2.19533 ± 0.00066	1.86976 ± 0.00056

4 Flux, geometry integration and stopping model

4.1 Incident spectrum

We retain the sea-level spectrum and normalization used in the original note. The geometric reference uses $I(\theta) = 70 \cos^2 \theta$ in $\text{m}^{-2} \text{s}^{-1} \text{sr}^{-1}$, the approximate vertical intensity above $p = 1 \text{ GeV}/c$ quoted by the PDG [1]. This reference tests acceptance without attenuation; it is not the energy-integrated rate used in the main tables.

The main rates integrate the modified Gaisser spectrum of Guan et al. [2], with total muon energy E in GeV:

$$j(E, \theta) = 10^4 C 0.14 \left[E + \frac{3.64}{(\cos \theta^*)^{1.29}} \right]^{-2.7} \left[\frac{1}{1 + 1.1E \cos \theta^*/115} + \frac{0.054}{1 + 1.1E \cos \theta^*/850} \right]. \quad (1)$$

Here j is in $\text{m}^{-2} \text{s}^{-1} \text{sr}^{-1} \text{GeV}^{-1}$; the factor 10^4 converts the published area unit. The curvature correction is

$$\cos \theta^* = \left[\frac{c^2 + p_1^2 + p_2 c^{p_3} + p_4 c^{p_5}}{1 + p_1^2 + p_2 + p_4} \right]^{1/2}, \quad c = \cos \theta, \quad (2)$$

with $(p_1, p_2, p_3, p_4, p_5) = (0.102573, -0.068287, 0.958633, 0.0407253, 0.817285)$. We set $C = 1.159538$ so that the vertical integral above $E = \sqrt{(1 \text{ GeV})^2 + m_\mu^2}$ is 70 in the stated intensity units. Both muon charges are included; electromagnetic shower particles and hadrons are outside the calculation.

4.2 Geometric acceptance

For two required anchor layers separated by D , let $\mathbf{a}$ and $\mathbf{b}$ be the ray positions on their horizontal centre planes. The slopes are $\mathbf{u} = (\mathbf{b} - \mathbf{a})/D$ and $c = (1 + |\mathbf{u}|^2)^{-1/2}$. The rate is

$$R = \int d^2a d^2b \frac{c^4}{D^2} F(E_{\min}, c) \mathbf{1}_c, \quad F(E_{\min}, c) = \int_{E_{\min}}^{\infty} j(E, c) dE. \quad (3)$$

The four powers of c include the projected area and solid-angle Jacobian. The geometric reference therefore has a $70c^6$ weight. Anchor rectangles are expanded to include tracks that enter a finite-thickness volume through its side. Each selection uses two layers that it actually requires, chosen to give efficient sampling. HCAL-only selections use broader anchors because the original TS-LYSO sampling region excludes most of their accepted trajectories.

For closely spaced HCAL layers and the any-layer triggers, directions are sampled with $q(c) = 4c^3$ and uniform azimuth. At each direction we construct a rectangle containing the projected acceptance and sample the transverse position within it. Exact prism intersections then retain or reject the ray. The geometric-reference weight is $70(\pi/2)A$ for rectangle area A in square metres. Boxes, rotated tracker rectangles and ECAL hexagons retain their finite thickness and segmentation. Adjacent HCAL bars are merged only after checking that they tile a rectangle without gaps.

Each of the 39 anchor families uses eight independently scrambled Sobol sequences with 2^{17} points per scramble, for 40,894,464 trial rays in the equipment study. The additional HCAL any-layer and individual-layer calculations use the same number of points per trigger definition. The supplied scripts record seeds, per-scramble estimates and reusable checkpoints. Numerical errors are the standard error across scrambles, not counting-statistics errors for a future data sample.

The tracker-threshold tables use exact set identities. If S_j is the sum of rates requiring each

specified j -layer tracker subset (with all other cuts fixed), then

$$R(N_T \geq k) = \sum_{j=k}^4 (-1)^{j-k} \binom{j-1}{k-1} S_j, \quad k \geq 1, \quad (4)$$

$$R(N_T = k) = \sum_{j=k}^4 (-1)^{j-k} \binom{j}{k} S_j. \quad (5)$$

For $k = 0$, the inclusive rate is S_0 . These formulas count the union without double counting. They are applied separately to every scramble to retain the covariance of estimates sharing a sample. Very small exclusive rates can be unresolved differences of larger numbers; signed estimates and numerical errors are retained in the CSV instead of being silently clipped to zero.

4.3 Material and the concrete slab

For each ray, the material column $X_d = \sum_i \rho_i \ell_i$ is integrated from above the stand to the lower surface of its last required layer. All installed devices contribute when crossed, whether or not their hits are required. For an any-layer HCAL station, the earliest crossed layer satisfies that station; the last such required station completion sets the stopping plane. The tracker silicon parents supply the full sensor material, including inactive borders; their same-material active daughters are not counted a second time.

The detector column is represented by the polystyrene CSDA range [4]. This remains an approximation to the mixed materials even though the chords and densities are evaluated ray by ray. Let $\mathcal{R}_p$ and $\mathcal{R}_c$ denote the polystyrene and concrete range functions. With kinetic energy in GeV,

$$K_d = \mathcal{R}_p^{-1}[\mathcal{R}_p(0.010) + X_d], \quad (6)$$

$$K_s = \mathcal{R}_c^{-1}[\mathcal{R}_c(K_d) + X_c/c], \quad E_{\min} = K_s + m_\mu. \quad (7)$$

The column X_d already includes the oblique path length, so no further $1/c$ factor is applied to it. Without concrete, $K_s = K_d$. The 10 MeV residual energy is the lower range-table reliability convention; it is not a detector threshold. The stopping plane is the lower Z face even for a track whose active chord ends on a side face.

The slab is 91.44 cm thick with density 2.30 g cm^{-3} , giving $X_c = 210.312 \text{ g cm}^{-2} = 2.10312 \text{ m}$ water equivalent [3]. It is horizontal and covers every accepted direction. Roof height, gaps, walls and local building structure have not been surveyed into this model. Unlike the earlier fixed-column comparison, a broad HCAL ray is not assigned the material budget of a golden event, and it need not emerge from equipment below the final required hit.

5 Checks and limits of the estimate

The new prism code reproduces every active-hit decision for the 192 saved centred-layout ROOT rays and a further 192 broad-angle ROOT rays (9,984 layer decisions in total). It also agrees with the independent Python intersection code for 16,384 broad rays. Material columns agree with ROOT to less than $2.4 \times 10^{-9} \text{ g cm}^{-2}$, including partial columns to the layer exit planes. All 896 specified-subset rates obey requirement monotonicity, as do the 280 inclusive multiplicity rows.

Independent pseudorandom estimates agree with the golden and strict ten-layer HCAL rates within 0.20 and 1.57 combined numerical standard errors, respectively. Separate angular checks of the three-station any-layer trigger and the union of all HCAL layers agree within 1.37 and 0.16 standard errors.

The tracker plateau also has a geometric check: the full TS–LYSO anchor envelope lies at least 6.62 mm inside every tracker active polygon.

The golden geometric reference is 1.55021 ± 0.00045 per hour, consistent with the earlier 1.54995 ± 0.00030 result. An independent angular proposal $q(c) = 2c$, which samples more oblique rays, agrees with all three strict single-station rates within 1.95 combined numerical standard errors; the largest difference is 0.0092%. The flux interpolation check has maximum relative residual 0.0209% over 250 test points; the energy-quadrature check against adaptive integration reaches 1.3×10^{-7} . These residuals are separate from the reported Sobol standard errors. Validation scripts and machine-readable checks are included in the reproduction package.

Table 5: Concrete rates per hour when the installed detector column on every ray is rescaled. The roof and hit logic are unchanged. The last column gives the no-roof rate-weighted mean material column in g cm^{-2} . These variations are not systematic uncertainty bounds.

Selection	$X_d/2$	Nominal X_d	$2X_d$	Mean X_d
Golden	1.868	1.842	1.793	32.75
All ten HCAL layers	1.142×10^4	1.129×10^4	1.105×10^4	27.11

The numerical uncertainty of the integration is much smaller than the physical uncertainty in the rate model. The flux normalization is not a local measurement: the PDG review notes measurements 10–15% below the reference value [1]. A change of normalization rescales all rates directly. Multiple scattering, stochastic energy loss and muon decay in the stand are omitted. Scattering can move tracks into or out of the narrow acceptance, so no one-sided correction is inferred here. The broad HCAL trigger and the golden sample need not have the same correction.

No light-yield, energy-deposition or charge threshold is imposed. In particular, a grazing active-volume chord is a hit in this model. In the earlier aligned calculation, requiring every layer midplane reduced the geometric reference from 1.550 to 1.278 per hour. That difference is a useful illustration of threshold sensitivity, not a calibrated inefficiency correction. The material-column variations in Table 5 likewise test one model ingredient and do not define a total uncertainty interval.

For a future measured coincidence, a schematic response relation is

$$R_{\text{recorded}} = f_{\text{live}} \int d\Gamma \frac{dR_\mu}{d\Gamma} P(\text{trigger and selection} \mid \Gamma) + R_{\text{accidental}} + R_{\text{other}}, \quad (8)$$

where Γ includes the trajectory and energy. For independent tracker layers with equal efficiency ϵ , a track crossing all four sensors has four-hit probability ϵ^4 and at-least-three-hit probability $\epsilon^4 + 4\epsilon^3(1-\epsilon)$. No efficiency value is assumed here. Correlations, dead areas and path-dependent response should be measured before using this relation quantitatively.

The geometry combines the STEP placement framework, ESA `ldmx-reduced-v3` central-detector internals and HCAL dimensions from the test-stand branch. The 12-bar and 8-bar HCAL counts and the registration remain inferred hardware inputs. In particular, the tracker uses the native 48×78 mm active sensors, and the ECAL has four active silicon layers; the CAD board and support counts do not directly specify these sensitive volumes. Agreement with the assembled hardware must therefore be checked when using these rates to plan measurements. The full CAD export omitted 965 placements and used placeholder material for unassigned CAD parts; the rate calculation uses the explicit native-material snapshot, not those placeholder supports. Prior GDML-schema checks and ROOT closure establish readable solids, not Geant4 transport, mechanical clearances or sensitive-detector response. No new native `ldmx-sw` transport run is claimed in this update.

6 Reproducing the study

The report directory contains the original geometry provenance, saved earlier studies and the new calculation in `combinations_20260919/`. From the report directory, run

```
PYTHON=/path/to/python3 bash build_report.sh
```

Python requires NumPy, SciPy and Matplotlib; a C++17 compiler builds the small prism-intersection library on first use. Tectonic builds the PDF. ROOT is used only by the additional geometry-validation script, not by the rate integrator. The default calculation uses eight scrambles of 2^{17} rays per anchor family; checkpoints allow an interrupted calculation to resume. The source files, input snapshots, numerical outputs and PDF are identified by SHA-256 manifests.

- `calculate_combinations.py`: all 896 specified subsets, 280 inclusive combinations and 280 exact tracker-count combinations.
- `calculate_hcal_triggers.py`: any-layer station coincidences, station unions and ten individual HCAL-layer singles.
- `ray_geometry.py` and `ray_geometry.cpp`: active hits and installed-material chords, with tracker material counted once.
- `results/*.csv`, `results/*.json` and `results/*replicates.npz`: rates, numerical errors, settings and per-scramble values. The labels and bit definitions are documented in `README.md`.

The previous PDF, source and build entry point are preserved under `previous_report/`. Rebuilding this update does not require rerunning the historical geometry variations or downloading range tables.

A Complete inclusive equipment combinations

Each column below specifies the required HCAL stations. Each row specifies the TS, LYSO and ECAL requirements (1 means required; 0 means unrestricted) and the minimum tracker count k . All layers in a selected HCAL station are required. Rates are per hour. The two tables contain the same 280 combinations for the two roof assumptions; numerical errors and full precision are in `inclusive_combinations.csv`. Omitted devices remain installed.

Table 6: 3 ft concrete: all inclusive equipment combinations, in muons/hour. A suffix k means 10^3 events/hour. Read the first column as S, Y, E and the minimum tracker count k .

S	Y	E	k	H_T	H_M	H_B	H_TH_M	H_TH_B	H_MH_B	$H_TH_MH_B$
1	1	1	4	1.859	1.859	1.842	1.859	1.842	1.842	1.842
1	1	1	3	1.859	1.859	1.842	1.859	1.842	1.842	1.842
1	1	1	2	1.859	1.859	1.842	1.859	1.842	1.842	1.842
1	1	1	1	1.859	1.859	1.842	1.859	1.842	1.842	1.842
1	1	1	0	1.859	1.859	1.842	1.859	1.842	1.842	1.842
1	1	0	4	1.911	1.897	1.870	1.897	1.870	1.870	1.870
1	1	0	3	1.911	1.897	1.870	1.897	1.870	1.870	1.870
1	1	0	2	1.911	1.897	1.870	1.897	1.870	1.870	1.870
1	1	0	1	1.911	1.897	1.870	1.897	1.870	1.870	1.870
1	1	0	0	1.911	1.897	1.870	1.897	1.870	1.870	1.870
1	0	1	4	2.567	2.567	2.545	2.567	2.545	2.545	2.545
1	0	1	3	2.567	2.567	2.545	2.567	2.545	2.545	2.545
1	0	1	2	2.567	2.567	2.545	2.567	2.545	2.545	2.545
1	0	1	1	2.567	2.567	2.545	2.567	2.545	2.545	2.545
1	0	1	0	2.567	2.567	2.545	2.567	2.545	2.545	2.545
1	0	0	4	2.699	2.678	2.639	2.678	2.639	2.639	2.639
1	0	0	3	2.699	2.678	2.639	2.678	2.639	2.639	2.639
1	0	0	2	2.699	2.678	2.639	2.678	2.639	2.639	2.639
1	0	0	1	2.699	2.678	2.639	2.678	2.639	2.639	2.639
1	0	0	0	2.699	2.678	2.639	2.678	2.639	2.639	2.639
0	1	1	4	16.704	16.704	16.557	16.704	16.557	16.557	16.557
0	1	1	3	17.912	17.912	17.755	17.912	17.755	17.755	17.755
0	1	1	2	25.146	25.146	24.926	25.146	24.926	24.926	24.926
0	1	1	1	26.339	26.339	26.108	26.339	26.108	26.108	26.108
0	1	1	0	31.508	31.508	31.232	31.508	31.232	31.232	31.232
0	1	0	4	24.970	24.785	24.456	24.785	24.456	24.456	24.456
0	1	0	3	27.831	27.625	27.262	27.625	27.262	27.262	27.262
0	1	0	2	68.225	67.810	63.164	67.694	63.164	63.164	63.164
0	1	0	1	76.403	76.071	70.029	75.717	70.029	70.029	70.029
0	1	0	0	249.1	367.6	179.0	240.9	170.2	179.0	170.2
0	0	1	4	37.149	37.149	36.824	37.149	36.824	36.824	36.824
0	0	1	3	40.913	40.913	40.554	40.913	40.554	40.554	40.554
0	0	1	2	55.565	55.565	55.078	55.565	55.078	55.078	55.078
0	0	1	1	60.989	60.989	60.455	60.989	60.455	60.455	60.455
0	0	1	0	1480.5	6244.5	6977.9	1480.5	1467.6	5945.9	1467.6
0	0	0	4	269.0	224.5	177.2	224.5	177.2	177.2	177.2
0	0	0	3	334.3	266.3	204.0	266.3	204.0	204.0	204.0
0	0	0	2	1543.0	759.4	454.0	733.3	454.0	454.0	454.0
0	0	0	1	1744.7	842.0	501.0	808.9	501.0	501.0	501.0
0	0	0	0	175.4k	69.5k	160.7k	14.0k	15.1k	32.5k	11.3k

Table 7: No overhead slab: all inclusive equipment combinations, in muons/hour. A suffix k means 10^3 events/hour. Read the first column as S, Y, E and the minimum tracker count k .

S	Y	E	k	H_T	H_M	H_B	H_TH_M	H_TH_B	H_MH_B	$H_TH_MH_B$
1	1	1	4	2.187	2.187	2.163	2.187	2.163	2.163	2.163
1	1	1	3	2.187	2.187	2.163	2.187	2.163	2.163	2.163
1	1	1	2	2.187	2.187	2.163	2.187	2.163	2.163	2.163
1	1	1	1	2.187	2.187	2.163	2.187	2.163	2.163	2.163
1	1	1	0	2.187	2.187	2.163	2.187	2.163	2.163	2.163
1	1	0	4	2.262	2.236	2.195	2.236	2.195	2.195	2.195
1	1	0	3	2.262	2.236	2.195	2.236	2.195	2.195	2.195
1	1	0	2	2.262	2.236	2.195	2.236	2.195	2.195	2.195
1	1	0	1	2.262	2.236	2.195	2.236	2.195	2.195	2.195
1	1	0	0	2.262	2.236	2.195	2.236	2.195	2.195	2.195
1	0	1	4	3.021	3.021	2.988	3.021	2.988	2.988	2.988
1	0	1	3	3.021	3.021	2.988	3.021	2.988	2.988	2.988
1	0	1	2	3.021	3.021	2.988	3.021	2.988	2.988	2.988
1	0	1	1	3.021	3.021	2.988	3.021	2.988	2.988	2.988
1	0	1	0	3.021	3.021	2.988	3.021	2.988	2.988	2.988
1	0	0	4	3.197	3.157	3.099	3.157	3.099	3.099	3.099
1	0	0	3	3.197	3.157	3.099	3.157	3.099	3.099	3.099
1	0	0	2	3.197	3.157	3.099	3.157	3.099	3.099	3.099
1	0	0	1	3.197	3.157	3.099	3.157	3.099	3.099	3.099
1	0	0	0	3.197	3.157	3.099	3.157	3.099	3.099	3.099
0	1	1	4	19.655	19.655	19.441	19.655	19.441	19.441	19.441
0	1	1	3	21.077	21.077	20.847	21.077	20.847	20.847	20.847
0	1	1	2	29.589	29.589	29.266	29.589	29.266	29.266	29.266
0	1	1	1	30.992	30.992	30.654	30.992	30.654	30.654	30.654
0	1	1	0	37.074	37.074	36.669	37.074	36.669	36.669	36.669
0	1	0	4	29.560	29.217	28.718	29.217	28.718	28.718	28.718
0	1	0	3	32.947	32.565	32.012	32.565	32.012	32.012	32.012
0	1	0	2	80.728	79.898	74.164	79.761	74.164	74.164	74.164
0	1	0	1	90.399	89.625	82.222	89.208	82.222	82.222	82.222
0	1	0	0	294.4	432.1	210.0	283.4	199.8	210.0	199.8
0	0	1	4	43.720	43.720	43.241	43.720	43.241	43.241	43.241
0	0	1	3	48.150	48.150	47.622	48.150	47.622	47.622	47.622
0	0	1	2	65.393	65.393	64.676	65.393	64.676	64.676	64.676
0	0	1	1	71.778	71.778	70.990	71.778	70.990	70.990	70.990
0	0	1	0	1741.7	7322.9	8160.1	1741.7	1722.6	6959.6	1722.6
0	0	0	4	318.2	264.5	208.1	264.5	208.1	208.1	208.1
0	0	0	3	395.4	313.7	239.5	313.7	239.5	239.5	239.5
0	0	0	2	1820.8	894.0	533.0	863.4	533.0	533.0	533.0
0	0	0	1	2058.7	991.1	588.3	952.3	588.3	588.3	588.3
0	0	0	0	205.8k	81.3k	187.8k	16.5k	17.7k	38.1k	13.3k

B Individual HCAL-layer singles

These rows require only the indicated active HCAL layer, with no requirements on other devices. They are physical muon-crossing estimates, not electronic singles. Layers are numbered from highest to lowest within each station.

Table 8: Single-layer muon crossings per hour. A suffix k means 10^3 events/hour. Other layer hits are unrestricted.

HCAL layer	No roof	3 ft concrete
H_T , layer 1	680.6k	579.1k
H_T , layer 2	679.9k	578.8k
H_M , layer 1	456.4k	388.8k
H_M , layer 2	456.2k	388.7k
H_M , layer 3	454.8k	388.0k
H_M , layer 4	454.6k	387.9k
H_B , layer 1	676.3k	577.0k
H_B , layer 2	675.8k	576.7k
H_B , layer 3	674.1k	575.8k
H_B , layer 4	673.7k	575.5k

C Earlier placement and shielding studies

The preceding report studied the offset ECAL, a 200 mm upward translation of the TS/tracker/LYSO assembly, ECAL centring, and physical ECAL removal. Table 9 preserves those results under their original fixed-column assumptions. They are useful for identifying the effect of placement; the new tables use the aligned layout and trajectory-dependent columns described in Section 4.

Table 9: Previously published placement comparisons (18 September). These rates use the earlier mean vertical column; they are preserved, not refitted.

Layout	Required layers	No roof/hour	Concrete/hour
Original, offset ECAL	26	1.407	1.199
TS/tracker/LYSO raised 200 mm	26	1.264	1.077
Raised assembly, centred ECAL	26	1.949	1.660
ECAL centred only	26	2.163	1.842
ECAL physically removed	22	2.214	1.883

Raising the central assembly alone left its 383.4 mm TS–LYSO separation unchanged but increased the distance to the offset ECAL, reducing the rate by 10.2%. Centring ECAL in that raised layout recovered acceptance, while centring it at the original heights gave a further 11.0% increase. Thus the vertical move was not needed to obtain the centring benefit.

In the original offset layout, the geometric factor was $G_2 = 4.0021 \times 10^{-6} \text{ m}^2 \text{ sr}$ and the geometric reference rate was 1.0085 ± 0.0004 per hour. Under that angular model, 90% and 99% of accepted rays lay within 4.82° and 6.28° of vertical. Requiring all layer midplanes instead of nonzero volume chords gave 0.853 per hour. The mean surface kinetic-energy requirements in the continuous-loss model were

0.104 GeV without a roof and 0.470 GeV with concrete. The corresponding mean event intervals were 42.6 and 50.1 minutes.

The original 96-ray material sample had mean vertical column 32.70 g cm^{-2} , including about 20.6 g cm^{-2} of HCAL scintillator. Doubling that column reduced the concrete prediction from 1.20 to 1.17 per hour; increasing the final kinetic-energy floor to 100 MeV also gave 1.17 per hour. The roof reduced the nominal rate by about 15%. The old `concrete_scan.csv` and `rate_comparison.pdf` retain the full thickness scan and angular distributions.

Physical ECAL removal required 22 layers and reduced the calculated remaining column to 24.32 g cm^{-2} . At the original fixed column, simply dropping the ECAL hit requirement gave 2.195 and 1.869 per hour for no roof and concrete, respectively. Removing its material increased those rates to 2.214 and 1.883 per hour. These historical numbers should not be substituted for the current installed, aligned-ECAL selections.

Each earlier variation used eight Sobol scrambles of 2^{18} point pairs. The stored independent pseudorandom checks agreed with the corresponding Sobol estimates. ROOT agreed with all 4,992 layer decisions in each 192-ray 26-layer test, and with all 4,224 decisions for ECAL removal. The original STEP hash and the source and generated GDML hashes remain in `geometry/provenance.json`; the registered geometry snapshot records revision `eae30ac8f64f694fa99bbc8ed30c1a77fef05135`.

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